MATA22 Linear Algebra I: Lecture Notes Summary (Weeks 0 to 12)

Source: Prof. T. Kielstra, University of Toronto Scarborough (MAT A22, Winter 2026). Summaries of the lecture notes Weeks 0 to 12 and Homework 12, using only what the notes state. Weeks 2 to 12 were read from scanned pages, so a few passages are marked [unclear in scan].

Week	Topic
0	Review of matrices and functions: notation and proof techniques
1	Number systems, fields and vector spaces
2	Subspaces and linear combinations
3	Systems of equations, augmented matrices, REF and RREF
4	Bases and dimension
5	Sums and direct sums
6	Linear transformations, coordinate vectors, matrix representations
7	Kernel, image and inverse image
8	Injectivity, surjectivity and composition
9	Inverses of linear transformations
10	Determinants and change of bases
11	Change of basis; eigenvalues and eigenvectors
12	Diagonalizability
HW 12	Diagonalizability and applying polynomials

Week 0: A Review of Matrices and Functions: Notations and Proof Techniques

Big idea

Three standard vector spaces, $\mathbb{F}^n$, $M_{n\times k}(\mathbb{F})$ and $\mathcal{F}(\mathbb{F})$, are each defined by coordinatewise, entrywise or pointwise addition, scalar multiplication and equality. Proofs about them fix one generic coordinate, entry or input, so the argument covers every index at once.

Definitions

- $\mathbb{F}^n$: $\{(x_1, \dots, x_n) : x_i \in \mathbb{F}\}$; $x + y := (x_1 + y_1, \dots, x_n + y_n)$; $cx := (cx_1, \dots, cx_n)$; $x = y$ iff $x_i = y_i$ for all i.
- $M_{n\times k}(\mathbb{F})$: $n \times k$ matrices with entries in $\mathbb{F}$; $(A + B)_{ij} := A_{ij} + B_{ij}$; $(cA)_{ij} := cA_{ij}$; $A = B$ iff the dimensions agree and $A_{ij} = B_{ij}$ for all i, j.
- Transpose: for $A \in M_{n\times k}(\mathbb{F})$, $A^T \in M_{k\times n}(\mathbb{F})$ with $(A^T)_{ij} := A_{ji}$.
- $\mathcal{F}(\mathbb{F}) = \{f : \mathbb{F} \to \mathbb{F}\}$; $(f + g)(x) := f(x) + g(x)$; $(cf)(x) := c\,f(x)$; $f = g$ iff $f(x) = g(x)$ for all $x \in \mathbb{F}$.
- Composition: $(f \circ g)(x) := f(g(x))$.

Theorems and results

- For $x, y \in \mathbb{F}^n$ and $c, d \in \mathbb{F}$: $c(x+y) = cx+cy$ and $(c+d)x = cx+dx$. Proof idea: compare the ith entries and reduce to distributivity in $\mathbb{F}$.
- The same two identities hold in $M_{n\times k}(\mathbb{F})$ (compare entries (i, j)) and in $\mathcal{F}(\mathbb{F})$ (compare values at x).
- $(A + B)^T = A^T + B^T$ for $A, B \in M_{n\times k}(\mathbb{F})$. Proof idea: $(A + B)^T_{ij} = A_{ji} + B_{ji}$, which is the (i, j) entry of $A^T + B^T$.
- $(f + g) \circ h = f \circ h + g \circ h$. Proof idea: evaluate both sides at an arbitrary x.

Techniques / procedures

1. Vector or matrix equality: check that dimensions match, then compare every coordinate or entry.
2. Function equality: fix an arbitrary input x and show the outputs agree.
3. Generic element: fix an arbitrary index i (or i, j, or x), prove the identity for that one entry, and conclude for all.
4. Transpose: swap row and column indices.

Worked examples in the notes

- $x = (1, 0, -2)$, $c = 3$, $y = \left(\frac{1}{3}, 0, -\frac{2}{3}\right) \to$ check $(cy)_i = c\,y_i = x_i$ for each $i \to x = cy$.
- $x = (1, 2, 3)$, $z = (1, 2) \to$ dimensions differ $\to x + z$ is not defined.
- $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} \to$ add entrywise $\to A + B = \begin{pmatrix} 6 & 8 \\ 10 & 12 \end{pmatrix}$.
- $A = \begin{pmatrix} 1 & -2 & 0 \\ 3 & 4 & 5 \end{pmatrix} \to (A^T)_{ij} = A_{ji} \to A^T = \begin{pmatrix} 1 & 3 \\ -2 & 4 \\ 0 & 5 \end{pmatrix}$.
- $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 0 \\ -1 & 2 \\ 3 & 1 \end{pmatrix} \to A^T + B$ is 3×2 and defined $\to (A^T + B)^T = \begin{pmatrix} 2 & 1 & 6 \\ 4 & 7 & 7 \end{pmatrix}$.
- $f(x) = 2x$, $g(x) = x$, $c = 2 \to (cg)(x) = 2x = f(x)$ for all $x \to f = cg$.

Common pitfalls

- Addition is defined only for objects of the same dimension: $x + z$ with $x \in \mathbb{F}^3$, $z \in \mathbb{F}^2$, and $A + C$ with A of size 2×2 and C of size 2×3, are undefined.
- $A^T + B^T$ is undefined when A^T is 3×2 and B^T is 2×3 (notes' example), even though $A + B^T$ is defined.
- Agreeing at one input does not make two functions equal; equality requires $f(x) = g(x)$ for all x.

Week 1: A Quick Tour of Number Systems, Fields and Vector Spaces

Big idea

Algebraic structures are built from axioms; this week fixes the fields $\mathbb{R}$ and $\mathbb{C}$ and the eight axioms of a vector space over a field. Most later verifications reduce to the fact that the scalars form a field.

Definitions

- Binary operation: a function $\boxplus : X \times X \to X$. Commutative: $x \boxplus y = y \boxplus x$. Associative: $x \boxplus (y \boxplus z) = (x \boxplus y) \boxplus z$.
- Field: a set $\mathbb{F}$ with addition $\boxplus$ and multiplication $\boxdot$ satisfying A1–A8: both operations associative and commutative; identities $\widehat{0}$ and 1; additive inverses; multiplicative inverses for $x \neq \widehat{0}$; and $a \boxdot (b \boxplus c) = a \boxdot b \boxplus a \boxdot c$.
- Complex number: $z = a + bi$ with $a, b \in \mathbb{R}$, $i = \sqrt{-1}$; $\mathrm{Re}(z) = a$, $\mathrm{Im}(z) = b$.
- Conjugate: $\bar{z} := a - bi$. Modulus: $|z| := \sqrt{\mathrm{Re}(z)^2 + \mathrm{Im}(z)^2}$.
- Complex operations: $(a + bi) + (c + di) := (a + c) + (b + d)i$; $(a + bi)(c + di) := (ac - bd) + (ad + bc)i$; $z_1 - z_2 := (a - c) + (b - d)i$.
- Division: $\dfrac{z_1}{z_2} := \dfrac{z_1 \bar{z}_2}{|z_2|^2}$.
- Vector space: a set V over a field $\mathbb{F}$, with $\boxplus : V \times V \to V$ and $\boxdot : \mathbb{F} \times V \to V$, satisfying A1–A8: $\boxplus$ associative and commutative; a zero vector 0; additive inverses; $c \boxdot (x \boxplus y) = (c \boxdot x) \boxplus (c \boxdot y)$; $(c + d) \boxdot x = (c \boxdot x) \boxplus (d \boxdot x)$; $(cd) \boxdot x = c \boxdot (d \boxdot x)$; $1 \boxdot x = x$.
- Examples: $\mathbb{R}^n$; $\mathcal{F}(\mathbb{R}) = \{f : \mathbb{R} \to \mathbb{R}\}$; $M_{m \times n}(\mathbb{R})$; over a general field, $\mathcal{F}(\mathbb{F})$ and $M_{m \times n}(\mathbb{F})$.

Theorems and results

- The zero vector is unique. Proof idea: $0' = 0 \boxplus 0' = 0' \boxplus 0 = 0$.
- $0\vec{z} = \vec{0}$ in every vector space. Proof idea: $0\vec{z} = (0 + 0)\vec{z}$, apply A6, then cancel.
- $(-1)\vec{x}$ is an additive inverse of $\vec{x}$. Proof: $\vec{x} \boxplus (-1)\vec{x} = (1 + (-1))\vec{x} = 0\vec{x} = \vec{0}$.
- A nonzero $z = a + bi$ has $z^{-1} = \dfrac{a}{a^2 + b^2} - \dfrac{b}{a^2 + b^2} i$.
- $\mathbb{C}$ is a field (stated as an exercise).

Techniques / procedures

1. Multiply complex numbers by expanding and using $i^2 = -1$.
2. Divide z_1 by z_2: multiply top and bottom by $\bar{z}_2$, then divide by $|z_2|^2$.
3. To write v as a linear combination, equate components to a linear system in the scalars and solve.
4. Verify a vector-space axiom coordinatewise, reducing each step to a field axiom.
5. Uniqueness proof: assume two objects have the property, then show they are equal.

Worked examples in the notes

- $(a + bi)(c + di) \to$ expand $\to (ac - bd) + (ad + bc)i$.
- $z \times z^{-1} \to$ multiply out $\to 1 + 0i$.
- $x = (1, 2, 3) + 2\big((1, 1, 1) - (3, 2, 1)\big) \to$ compute inside, scale, add $\to (-3, 0, 3)$.
- Is $(-1, 1) = a(1, 2) + b(1, 1)$? $\to -1 = a + b$, $1 = 2a + b \to (a, b) = (2, -3)$. (The printed check "$2(1, 2) - 2(1, 1)$" equals $(0, 2)$, not $(-1, 1)$; the solved values do check.)
- $c \boxdot (f \boxplus g) = c \boxdot f \boxplus c \boxdot g$ in $\mathcal{F}(\mathbb{R}) \to$ compare at arbitrary $x \to$ distributivity in $\mathbb{R} \to$ equal.

Common pitfalls

- Subtraction is neither commutative ($2-3 \neq 3-2$) nor associative ($(1-2)-3 \neq 1-(2-3)$).
- Division is not commutative ($\frac{3}{2} \neq \frac{2}{3}$); complex numbers are divided via $\bar{z}_2/|z_2|^2$.
- Exponents: $2^3 \neq 3^2$, and $(2^3)^4 \neq 2^{(3^4)}$ [bracketing unclear in scan].

Week 2: Subspaces and Linear Combinations

Big idea

A subspace is a subset of a vector space that is itself a vector space under the same operations, and closure under $cx+y$ tests this. Spans, linear combinations and linear independence describe subspaces and detect redundant vectors.

Definitions

- Subspace: $W \subseteq V$ that is a vector space with the same 0 and operations as V.
- Linear combination of $S \subseteq V$: $a_1v_1 + \cdots + a_kv_k$ with $v_i \in S$, $a_i \in \mathbb{F}$.
- $\text{span}(S)$: all linear combinations of elements of S; $\text{span}(\emptyset) = \{0\}$.
- Sum of subsets: $A + B = \{a + b : a \in A,\ b \in B\}$.
- Linearly dependent (L.D.): some constants $a_1, \dots, a_k$, not all zero, satisfy $\sum a_iv_i = 0$. Linearly independent (L.I.): not dependent.
- Redundant: $w \in S$ with $w \in \text{span}(S \setminus \{w\})$.
- $\text{col}(A)$, $\text{row}(A)$: spans of the columns and rows of A.

Theorems and results

- Closure characterization: a nonempty $W \subseteq V$ is a subspace iff $cx + y \in W$ for all $c \in \mathbb{F}$, $x, y \in W$. Proof idea: $0 \in W$ follows from $(-1)x + x$, and $(-1)x \in W$ gives inverses.
- $\text{span}(S)$ is a subspace of V (case split on $S = \emptyset$).
- If W_1, W_2 are subspaces, so are $W_1 \cap W_2$ and $W_1 + W_2$.
- If $W_i = \text{span}(S_i)$, then $W_1 + W_2 = \text{span}(S_1 \cup S_2)$.
- S is linearly dependent iff S contains a redundant vector. Proof idea: a relation with $a_n \neq 0$ solves for v_n; conversely $w = \sum a_iw_i$ gives $(-1)w + \sum a_iw_i = 0$.
- For $S = \{v_1, \dots, v_n\}$ the following are equivalent: (1) S is L.I.; (2) $0 \in \text{span}(S)$ is represented uniquely; (3) every vector of $\text{span}(S)$ is represented uniquely. Proof: $1 \Rightarrow 2 \Rightarrow 3 \Rightarrow 1$.

Techniques / procedures

1. Subspace test: show $W \neq \emptyset$ (e.g., $0 \in W$), then show $cx + y \in W$.
2. To show $W = \text{span}(S)$: solve the constraints, write a generic element as free parameters times fixed vectors, and prove both inclusions.
3. Independence test: set $\sum a_iv_i = 0$ and show only $a_i = 0$ works.
4. Remove redundant vectors to shrink a spanning set without changing its span.
5. Case-by-case proof: split into cases covering all possibilities and prove each.

Worked examples in the notes

- $\{(a, 0) : a \in \mathbb{R}\} \to c(a, 0) + (b, 0) = (ca + b, 0) \to$ subspace.
- Continuous functions in $\mathcal{F}(\mathbb{R}) \to cf + g$ is continuous $\to$ subspace.
- $\{(x, y, z) : x + y + z = 0\} \to$ substitute $x = -y - z \to y(-1, 1, 0) + z(-1, 0, 1) \to$ span of $\{(-1, 1, 0), (-1, 0, 1)\}$.
- $\{(x, 0)\} + \{(y, y)\} = \mathbb{R}^2 \to (a, b) = (a - b, 0) + (b, b) \to$ both inclusions.
- $\{(1, 2), (1, 0), (0, 1)\} \to (1, 2) = (1, 0) + 2(0, 1) \to (1, 2)$ is redundant.
- $\{(1, 2, 3), (1, 0, 1), (2, 2, 4)\} \to (1, 2, 3) + (1, 0, 1) = (2, 2, 4) \to$ dependent.
- $\{\sin^2 x, \cos^2 x, 1\} \to \sin^2 x + \cos^2 x - 1 = 0 \to$ dependent.
- $\{\sin x, \cos x\} \to x = 0$ gives $b = 0$; $x = \dfrac{\pi}{2}$ gives $a = 0 \to$ independent.

Common pitfalls

- A proof by cases must cover every case; examples where the statement holds are not a proof.
- Equality of sets needs both $A \subseteq B$ and $B \subseteq A$.
- The empty sum is 0, so $\text{span}(\emptyset) = \{0\}$, not $\emptyset$.
- A set is dependent if any finite subset of it is dependent.

Week 3: Systems of Linear Equations, Augmented Matrices, Row Echelon Form and Row Reduced Echelon Form

Big idea

Linear systems are encoded as augmented matrices and solved by elementary row operations, which preserve the solution set and produce REF or RREF. The solution set is empty, a single point, or an affine set $p + \text{span}(S)$.

Definitions

- Linear system: n equations in k variables. A solution is a k-tuple satisfying all equations simultaneously. Equivalent systems have the same solution set.
- Homogeneous: every right-hand constant is zero.
- Augmented matrix: $[A \mid b]$.
- Elementary row operations: $R_i + \lambda R_j \to R_i$ ($i \neq j$); $\lambda R_i \to R_i$ ($\lambda \neq 0$); $R_i \leftrightarrow R_j$.
- REF: (1) the first nonzero entry of each nonzero row is its pivot, in column $j(i)$; (2) zeros lie below each pivot; (3) $j(i) < j(i+1)$ for consecutive nonzero rows (staircase).
- RREF: REF, with every pivot equal to 1 and zeros above each pivot too.
- Basic variables: variables in pivot columns. Free variables: all others.
- Affine set: $p + \text{span}(S)$ for a fixed vector p.

Theorems and results

- Finitely many elementary row operations produce an equivalent system.
- Every linear system has a unique RREF (proof left as an exercise).
- A homogeneous system always has the trivial solution 0.
- If $n < k$, every homogeneous system with k variables and n equations has a nontrivial solution. Proof idea: basic variables number at most n, so at least $k - n \geq 1$ variables are free.
- If v_p solves $[A \mid b]$, the solution set is $\{v_p + \theta : \theta \text{ solves } [A \mid 0]\}$ (accepted as fact this week).
- A system has no solution, exactly one, or infinitely many.

Techniques / procedures

1. Gaussian elimination: for each column, bring a nonzero entry at or below the current row into the pivot position, then clear the entries below it.
2. Gauss–Jordan: as above, also scale each pivot to 1 and clear the entries above it.
3. Back substitution: solve the last nonzero equation first, with free variables as parameters s_i.
4. Particular plus homogeneous: reduce to $[A' \mid b']$; set free variables to 0 for v_p; solve $[A' \mid 0]$ for the homogeneous set.
5. Dependence test: put the vectors as columns of a homogeneous system; only the trivial solution means independent.
6. Writing W as a span: write the constraints as a homogeneous system, reduce to RREF, solve for basic variables, and factor out the free parameters.

Worked examples in the notes

- $\begin{pmatrix} 1 & 0 & 1 \\ 2 & -1 & 1 \\ 3 & 0 & 0 \end{pmatrix}$ (columns $(1,2,3), (0,-1,0), (1,1,0)$) $\to$ row reduce to I_3 $\to$ independent.
- Is $f = 2x + 2 \in \text{span}\{x^2 + 1,\ x^2 - 1\}$? $\to$ coefficient matching gives $c_1 + c_2 = 0$, $0 = 2$, $c_1 - c_2 = 2 \to$ inconsistent $\to f \notin$ span.

- $\left(\begin{array}{cccc|c} 1 & 2 & 0 & 3 & 3 \\ 2 & 4 & 0 & 6 & 6 \\ 1 & 1 & 4 & 3 & 5 \end{array}\right) \to$ Gauss–Jordan to $\left(\begin{array}{cccc|c} 1 & 0 & 8 & 3 & 7 \\ 0 & 1 & -4 & 0 & -2 \end{array}\right) \to (7,-2,0,0)+s_1(-8,4,1,0)+s_2(-3,0,0,1)$.
- $W = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a+2b+3c=0,\ a+b+c=0 \right\} \to$ RREF gives $a-c=0$, $b+2c=0 \to W = \text{span}\left\{ \begin{pmatrix} 1 & -2 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$.
- $a+b=3$, $b+2c+d=6$, $d=5 \to$ back substitution $\to (2,1,0,5)+s(2,-2,1,0)$.
- Is $(1,i) \in \text{span}\{(1,1),(1,2)\}$ over $\mathbb{C}$? (Target OCR-garbled; read as $(1,i)$.) $\to x_1+x_2=1$, $x_1+2x_2=i \to (2-i,\ i-1) \to$ yes.

Common pitfalls

- Swapping rows gives an equivalent system; swapping columns (variables) generally does not.
- The homogeneous solution set decides uniqueness; existence needs a particular solution.
- For $b \neq 0$, the graph of $y=mx+b$ is affine, not a subspace; for $b=0$ it is $\text{span}\{(1,m)\}$.
- Affine representations $p+\text{span}(S)$ are not unique.

Week 4: Bases and Dimension

Big idea

A basis is a linearly independent spanning set, and all finite bases of a space have the same size, so $\dim(V)$ is well defined. Spanning sets can therefore be shrunk to bases, and independent sets can be extended to bases.

Definitions

- Spanning set: $S \subseteq V$ with $V = \text{span}(S)$.
- Basis: $S \subseteq V$ that is linearly independent and spans V.
- Finite-dimensional: V admits a finite basis; $\dim(V)$ is the number of elements in any basis.
- Proper subspace: a subspace S with $S \neq V$.
- Standard basis of $\mathbb{R}^n$ or $\mathbb{C}^n$: $\{e_1, \dots, e_n\}$, where e_i has 1 in position i and 0 elsewhere.

Theorems and results

- S is a basis of V iff every vector of V has a unique representation in $\text{span}(S)$. Proof: spanning gives existence and independence gives uniqueness, and conversely.
- Extension lemma: for S linearly independent and $x \notin S$, $S \cup \{x\}$ is linearly independent iff $x \notin \text{span}(S)$. Forward direction by contradiction; backward direction by solving for x.
- If V has a finite spanning set T and S is a finite independent subset, a finite basis $S' \supseteq S$ exists. Proof idea: add each $y \in T$ that is not in the span of the current set.
- Replacement result: if $T = \{y_1, \dots, y_n\}$ spans V and $S = \{x_1, \dots, x_k\}$ is independent, the notes print "$n < k$" [unclear in scan]. The proof (writing each x_i in terms of T, then a homogeneous system with more unknowns than equations) supports $k \leq n$.
- Any two finite bases of V have the same number of elements.
- If $\dim(V) = n$ and S is independent with $|S| = n$, then S is a basis. Proof by contradiction via the extension lemma.
- Suggested exercise: for finite-dimensional V and a proper subspace S, $\dim(S) < \dim(V)$.

Techniques / procedures

1. To show S is a basis: (i) solve $\sum c_i v_i = 0$ and find only the trivial solution; (ii) solve $v = \sum a_i v_i$ for arbitrary v.
2. For $\mathbb{R}^n$, write $v = \sum v_i e_i$ to show the standard basis spans.
3. To extend an independent set, go through a finite spanning set T, adding each $y_j \notin$ span of the current set.
4. If an independent set has $\dim(V)$ elements, it is a basis.

Worked examples in the notes

- Standard basis of $\mathbb{R}^n \to$ check independence and $v = \sum v_i e_i \to$ basis.
- $\{(1,1),(1,-1)\}$ spans $\mathbb{R}^2 \to$ solve $a + b = x$, $a - b = y \to (a,b) = \left(\frac{x+y}{2}, \frac{x-y}{2}\right)$ [final expression garbled in scan; this is the solution of the system as written].
- $\{(1,1),(1,-1)\}$ is independent $\to a + b = 0$, $a - b = 0 \to a = b = 0 \to$ basis of $\mathbb{R}^2$.
- $\{1, a, \dots, a^n\} \to$ basis of $P_n(\mathbb{R}) \to n + 1$ elements, so dim $= n + 1$.
- $\{a^k : k \in \mathbb{N}\} \to$ basis of $P(\mathbb{R}) \to$ infinite.
- $\mathcal{F}(\mathbb{R}) \to$ a basis exists only via the Axiom of Choice $\to$ no explicit basis.

Common pitfalls

- The notes single out recalling and applying previously proved theorems as the step students most often struggle with.
- $\dim(V)$ is defined only for finite-dimensional spaces; $P(\mathbb{R})$ needs an infinite basis.
- The backward direction of the extension lemma needs a different argument from the forward direction.

Week 5: Sums and Direct Sums

Big idea

For subspaces $U, V \subseteq W$, the sum $U + V$ need not have unique decompositions $w = u + v$. A direct sum is the case where every vector of $U + V$ has exactly one such decomposition, which the notes show is equivalent to $U \cap V = \{0\}$.

Definitions

- **Sum**: $U + V = \{u + v : u \in U,\ v \in V\}$.
- **Direct sum** ($U \oplus V$): U and V form a direct sum if $0 = u + v$ with $u \in U$, $v \in V$ forces $u = 0$ and $v = 0$.
- **Direct sum of** $U_1, \dots, U_k$: $0 = u_1 + \cdots + u_k$ with $u_i \in U_i$ forces $u_1 = \cdots = u_k = 0$.
- **Linearly independent family of subspaces**: the same condition as the direct sum of $U_1, \dots, U_k$.

Theorems and results

- **Two subspaces.** TFAE: (1) $U + V$ is direct; (2) every $w \in U + V$ is uniquely $w = u + v$; (3) $U \cap V = \{0\}$. *Idea:* (1)⇒(2) subtract two representations to get a zero representation; (2)⇒(3) write $0 = x + (-x)$ for $x \in U \cap V$; (3)⇒(1) if $u = -v$ then $u \in U \cap V$.
- **Finitely many subspaces** (W finite-dimensional, $\mathcal{B}_i$ a basis of U_i). TFAE: (1) the zero vector has a unique representation; (2) every vector of $U_1 + \cdots + U_k$ is uniquely represented; (3) $\mathcal{B}_1 \cup \cdots \cup \mathcal{B}_k$ is linearly independent. (Proof of (1)⇔(2) left as exercise.)
- **Dimension criterion.** TFAE: (1) direct sum; (2) $\mathcal{B}_1 \cup \cdots \cup \mathcal{B}_k$ is a basis of $W_1 \oplus \cdots \oplus W_k$; (3) $\dim(W_1 + \cdots + W_k) = \dim W_1 + \cdots + \dim W_k$. *Idea:* (1)⇒(2) spanning by grouping, independence from uniqueness of zero; (2)⇒(3) count elements; (3)⇒(1) the union spans, equal dimension makes it a basis, so the zero representation is unique.
- **Pairwise is not enough.** Pairwise direct sums do not imply a direct sum of all the subspaces.

Techniques / procedures

1. To show $U + V$ is direct: take $x \in U \cap V$, write $0 = x + (-x)$, conclude $x = 0$, so $U \cap V = \{0\}$.
2. To show $W = U + V$: take arbitrary $w \in W$ and exhibit $u \in U$, $v \in V$ with $w = u + v$.
3. Even/odd split: $f_e(x) = \dfrac{f(x) + f(-x)}{2}$, $f_o(x) = \dfrac{f(x) - f(-x)}{2}$; check f_e is even, f_o is odd, and $f = f_e + f_o$.
4. For three or more subspaces, test the zero vector against all of them simultaneously; pairwise checks are not sufficient.
5. In finite dimension, compare $\dim(W_1 + \cdots + W_k)$ with $\sum \dim W_i$.

Worked examples in the notes

- $U = V = \text{span}\{(1,0)\} \subseteq \mathbb{R}^2 \to (1,0) = 1(1,0) + 0(1,0)$ and a second split with $u = [\text{unclear in scan}]\,(1,0)$, $v = 3(1,0) \to$ two representations, so not direct.
- $W_1 = \text{span}\{e_1, e_2\}$ (scan reads "e_s, e_p") and $W_2 = \text{span}\{e_3\}$ in $\mathbb{R}^3 \to e_3 \notin W_1 \to \mathbb{R}^3 = W_1 \oplus W_2$.
- $\text{span}\{(1,0)\},\ \text{span}\{(0,1)\},\ \text{span}\{(1,1)\}$ in $\mathbb{R}^2 \to 0 = 1(1,0) + 1(0,1) + (-1)(1,1)$ with nonzero coefficients → pairwise direct, but not a direct sum.
- Even and odd functions on $\mathbb{R} \to f = f_e + f_o$ by the formulas above; if $g \in E \cap O$ then $2g(x) = 0 \to \mathbb{R}^{\mathbb{R}} = E \oplus O$.

- $M_n(\mathbb{R}) = S \oplus K$ (symmetric and skew-symmetric) $\rightarrow$ suggested exercise only; hint $\frac{1}{2}(A + A^T)$ and $\frac{1}{2}(A - A^T)$.

Common pitfalls

- $U \cap V = \{0\}$ does not mean vectors chosen from U and V are automatically linearly independent; independence is a property of a set of vectors.
- Representations $w = u + v$ in $U + V$ need not be unique in general.
- Pairwise directness does not guarantee global directness; the zero vector must have a unique representation using all subspaces at once.

Week 6: Linear Transformations, Coordinate Vectors and Matrix Representations

Big idea

A linear transformation $T : V \to W$ is a map that preserves addition and scalar multiplication. Once bases are fixed, T is encoded by a matrix, so computing $T(v)$ becomes multiplying coordinate vectors by that matrix.

Definitions

- **Linear transformation**: $T(u+v) = T(u) + T(v)$ and $T(av) = aT(v)$ for all $u, v \in V$, $a \in F$ (operations of V on the left, of W on the right).
- **Coordinate vector**: for a basis $\alpha = \{v_1, \dots, v_n\}$ and $v = \sum_i b_i v_i$, $[v]_\alpha = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$.
- **Matrix of** T: for bases α of V and β of W, $[T]_\alpha^\beta$ has i-th column $[T(v_i)]_\beta$.
- **Standard inner product**: $\langle u, v\rangle = \sum_i u_i v_i$; **length** $\|v\| = \sqrt{\langle v, v\rangle}$.
- **Orthogonal**: nonzero u, v with $\langle u, v\rangle = 0$.
- **Projection onto** $L = \operatorname{span}\{a\}$: $P_a(v) = \dfrac{\langle a, v\rangle}{\langle a, a\rangle}\, a$.

Theorems and results

- For linear T: $T(0_V) = 0_W$. *Idea:* $T(0_V) = T(0 \cdot 0_V) = 0 \cdot T(0_V) = 0_W$.
- T is linear iff $T(au + bv) = aT(u) + bT(v)$ for all $u, v \in V$, $a, b \in F$.
- The zero and identity transformations are linear.
- If $S, T : V \to W$ are linear and agree on a basis, then $S = T$. *Idea:* write $v = \sum a_i v_i$ and apply linearity to both maps.
- A linear $T : V \to W$ with $\dim V = n$, $\dim W = k$ is uniquely determined by nk scalars.
- For nonzero $u, v \in \mathbb{R}^n$: $\cos\theta = \dfrac{\langle u, v\rangle}{\|u\|\,\|v\|}$. *Idea:* expand $\|u - v\|^2$ with the inner product and compare with the law of cosines. Hence $u \perp v$ iff $\langle u, v\rangle = 0$.
- **Coordinate formula**: $[T(v)]_\beta = [T]_\alpha^\beta [v]_\alpha$. *Idea:* expand $v = \sum_j a_j v_j$, apply linearity, and reorder finite sums.
- **Fundamental theorem** (proof postponed to the textbook): for $M \in M_{k\times n}(\mathbb{R})$, $T_M(v) = w \iff M[v]_\alpha = [w]_\beta$ defines a linear map; every linear T satisfies $T = T_{[T]}$, and $M = [T_M]$.

Techniques / procedures

1. Prove linearity: check additivity and homogeneity separately, or check $T(au + bv) = aT(u) + bT(v)$ directly.
2. Build $[T]_\alpha^\beta$: compute each $T(v_i)$, write it in basis β, and place its coordinates in column i.
3. Find $[v]_\alpha$: solve $v = \sum_i b_i v_i$; the b_i are the entries.
4. Compute $[T(v)]_\beta$ as $[T]_\alpha^\beta [v]_\alpha$: multiply each row of $[T]$ by the column $[v]_\alpha$.
5. Projection: solve $\langle a,\, v - ca\rangle = 0$ for c, giving $c = \dfrac{\langle a, v\rangle}{\langle a, a\rangle}$.
6. Rotation: $R_\theta(e_1) = (\cos\theta, \sin\theta)$ and $R_\theta(e_2) = (-\sin\theta, \cos\theta)$ are the columns.

Worked examples in the notes

- $T(x) = 7x$ on $\mathbb{R} \to$ check $T(u+v)$ and $T(au)$ separately using distributivity and commutativity $\to$ linear.
- $v = (2,0)$, α standard, $\beta = \{(1,1),(1,-1)\} \to [v]_\alpha = (2,0)^T$, $[v]_\beta = (1,1)^T$ (printed expansion garbled: [unclear in scan]).
- Identity from $\alpha = \{(0,1),(1,0)\}$ to $\beta = \{(1,1),(1,-1)\} \to$ express each α-vector in $\beta \to$ columns $(\frac{1}{2}, -\frac{1}{2})^T$ and $(\frac{1}{2}, \frac{1}{2})^T$ (scan partly unclear).
- Rotation R_θ on $\mathbb{R}^2 \to$ columns from step 6 $\to [R_\theta] = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ (printed matrix partly unclear; columns legible).
- Projection of $(4,3)$ onto span$\{(1,0)\} \to \dfrac{\langle (1,0),(4,3)\rangle}{\langle (1,0),(1,0)\rangle}(1,0) \to (4,0)$.
- $T(x,y,z) = (x{+}y{+}z,\ y{+}z,\ z)$ on $\mathbb{R}^3 \to$ columns $[T(e_i)] \to$ check $[T(1,2,3)] = [T]\,[(1,2,3)]$ $\to$ notes report agreement (some entries [unclear in scan]).
- $M = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 1 & 2 & 3 \end{pmatrix}$ from $P_2(\mathbb{R})$ (basis $1, x, x^2$) to $\mathbb{R}^3 \to T_M(a_1 + a_2x + a_3x^2) = (a_1 + 2a_2 + a_3,\ a_2,\ a_1 + 2a_2 + 3a_3)$.

Common pitfalls

- Writing $T(u+v) = T(u) + T(v)$ hides the fact that V and W may have different operations; track them separately.
- Coordinate vectors depend on the basis: $[v]_\alpha$ and $[v]_\beta$ are generally different.

Week 7: Kernel, Image and Inverse Image

Big idea

The kernel, image and inverse image are natural subspaces attached to a linear map, and in finite dimensions they are computed by solving linear systems for the matrix representation. Rank-nullity links their dimensions: $\dim\ker T + \dim \text{image}\, T = \dim V$.

Definitions

- **Kernel**: $\ker(T) = \{v \in V : T(v) = 0_W\}$.
- **Nullspace**: $\text{nullspace}(A) = \{x : Ax = 0\}$.
- **Image**: $\text{image}(T) = \{w \in W : w = T(v) \text{ for some } v \in V\}$.
- **Column space**: $\text{col}(A) = \text{span}\{a_1, \dots, a_n\}$, where a_i are the columns of A.
- **Inverse image**: $T^{-1}(S) = \{v \in V : T(v) \in S\}$; $T^{-1}(w)$ means $T^{-1}(\{w\})$.

Theorems and results

- $\ker(T)$ and $\text{image}(T)$ are subspaces (closure test). *Idea:* linearity gives $T(x+y) = 0$ and $T(cx) = 0$ for $x, y \in \ker T$; similarly for images.
- **Matrix form**: $v \in \ker T \iff [T]_\alpha^\beta [v]_\alpha = [0]_\beta$, so $\ker T$ corresponds to $\text{nullspace}([T]_\alpha^\beta)$.
- If $\{v_i\}$ spans V, then $\{T(v_i)\}$ spans $\text{image}(T)$. *Idea:* write $v = \sum a_i v_i$ and apply linearity.
- $\{[w_i]_\beta\}$ is a basis of $\text{col}([T]_\alpha^\beta)$ iff $\{w_i\}$ is a basis of $\text{image}(T)$ (proof left as exercise).
- Pivot columns of A (the original columns, at positions of pivots in RREF) form a basis of $\text{col}(A)$ (proof left as exercise).
- **Rank-nullity**: $\dim\ker T + \dim \text{image}\, T = \dim V$. *Idea:* extend a basis of $\ker T$ to a basis of V; the images of the added vectors form a basis of $\text{image}\, T$.
- If $Ax = b$ has a solution x_p, its full solution set is $x_p + \text{nullspace}(A)$. *Idea:* the difference of two solutions lies in the nullspace.
- The solution set of $Ax = b$ is a subspace iff $b = 0$. *Idea:* 0 in the set forces $b = 0$; if $b = 0$ the set is the nullspace.

Techniques / procedures

1. Show a set is a subspace: contains 0 and is closed under addition and scalar multiplication.
2. Kernel: form $[T]_\alpha^\beta$; row-reduce $[\,[T] \mid 0\,]$; set each free variable to a parameter; write the solution as a combination; convert (b_i) to $v = \sum b_i v_i$.
3. Image: take the pivot columns of $[T]$ and map them back to W through β.
4. Solve $T(v) = w$: row-reduce $[\,[T] \mid [w]_\beta\,]$; then $T^{-1}(w) = \{v_p\} + \ker T$.
5. Check with rank-nullity: free variables count $\dim\ker T$, pivots count $\dim \text{image}\, T$.

Worked examples in the notes

- Projection onto the z-axis, $T_1 : \mathbb{R}^3 \to \mathbb{R}^3 \to \ker T_1 = \{(x, y, 0)\}$, $\text{image}\, T_1 = \{(0, 0, z)\} \to 2 + 1 = 3$.
- Rotation by $\pi/2$, $R(x, y) = (-y, x)$, on $\mathbb{R}^2 \to \ker = \{(0, 0)\}$, $\text{image} = \mathbb{R}^2 \to 0 + 2 = 2$.
- $T : M_{2\times 2}(\mathbb{R}) \to P_2(\mathbb{R})$ (formula partly [unclear in scan]) $\to$ reduced system $x_1 + x_4 = 0$, $x_2 - x_3 = 0 \to \ker T$ basis $\{-E_{11} + E_{22},\ E_{12} + E_{21}\}$.
- Differentiation $D : P_3(\mathbb{R}) \to P_2(\mathbb{R}) \to [D]$ has pivots in the columns for $x, x^2, x^3 \to$ image basis $\{1, 2x, 3x^2\}$, kernel basis $\{1\}$.
- $T(x, y) = (x - y,\ x - y)$ on $\mathbb{R}^2 \to \ker T = \text{span}\{(1, 1)\}$; $T^{-1}(2, 2) = (2, 0) + \text{span}\{(1, 1)\}$; $T^{-1}(2, 0) = \emptyset$ (inconsistent system).

- $T : P_2(\mathbb{R}) \to \mathbb{R}^3$, $T(p) = (p(0), p(1), p(2)) \to$ solve $[T][p] = [(1,2,4)] \to$ unique solution $p(x) = 1 + \frac{1}{2}x + \frac{1}{2}x^2$.
- $A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$, $b = (1,3)^T \to x_p = (-2,0,3)^T$, nullspace$(A) =$ span$\{(-1,1,0)^T\}$ (scan partly unclear).

Common pitfalls

- $T^{-1}(S)$ is not an inverse function: if $T(1,1) = T(2,2) = (0,0)$, then $T^{-1}(0,0)$ is not a single vector, and the notation does not imply an inverse map exists.
- The solution set of $Ax = b$ is a subspace only when $b = 0$.
- For a basis of the column space, use the columns of the original matrix that match RREF pivot columns, not the RREF columns themselves.

Week 8: Injectivity, Surjectivity and Composition

Big idea

For linear maps, injectivity is equivalent to a trivial kernel, and dimension counting (rank-nullity) strongly constrains injectivity and surjectivity. Composition of linear maps is linear, and its matrix is the product of the matrices.

Definitions

- **Injective**: $f(v_1) = f(v_2)$ implies $v_1 = v_2$.
- **Surjective**: for every $w \in W$ there is $v \in V$ with $f(v) = w$.
- **Sum and scaling of maps**: $(S+T)(v) = S(v) + T(v)$ and $(aS)(v) = a\,S(v)$.
- **Composition**: $(S \circ T)(x) = S(T(x))$, defined when $\text{image}(T) \subseteq \text{domain}(S)$.
- **Matrix product**: $p_{ij} = \sum_N t_{iN}\, s_{Nj}$ for $[T]\,[S]$.

Theorems and results

- **Finite sets.** For $f : D \to R$ with D, R finite: (1) $|D| < |R|$ implies f is not surjective; (2) $|D| > |R|$ implies f is not injective; (3) if $|D| = |R|$, f is injective iff surjective.
- **Injectivity.** For linear T, TFAE: T injective; $\ker T = \{0\}$; $\dim \ker T = 0$. *Idea:* $T(u) = T(v)$ gives $T(u-v) = 0$; conversely $T(v) = 0 = T(0)$ forces $v = 0$.
- **Surjectivity** (W finite-dimensional). For linear T, TFAE: T surjective; $\text{image}(T) = W$; $\dim \text{image}(T) = \dim W$. *Idea:* a subspace of W with dimension $\dim W$ equals W.
- If $\dim W < \dim V$ (finite), then T is not injective. *Idea:* rank-nullity gives $\dim \ker T \geq \dim V - \dim W > 0$.
- If $\dim V < \dim W$ (finite), then T is not surjective. *Idea:* $\dim \text{image}\, T \leq \dim V < \dim W$.
- If $\dim V = \dim W = n < \infty$, then T is injective iff surjective. *Idea:* rank-nullity gives $\dim \text{image}\, T = n - \dim \ker T$.
- For linear $S : U \to V$ and $T : V \to W$: $\ker S \subseteq \ker(TS)$ and $\text{image}(TS) \subseteq \text{image}(T)$.
- Composition of linear maps is linear; composition is associative; composition and addition distribute (partly proved in notes, rest left as exercise).
- **Matrix of a composition**: $[TS]_\alpha^\gamma = [T]_\beta^\gamma\,[S]_\alpha^\beta$. *Idea:* compare the (i,j) entries using uniqueness of coordinates and rearranging sums.
- Matrix multiplication is associative and distributive (proved entrywise).

Techniques / procedures

1. Injectivity: row-reduce $[T]\,v = 0$; if the only solution is trivial (a pivot in every column), T is injective.
2. Surjectivity: find pivot columns of $[T]$ to get $\text{image}(T)$, or compare $\dim \text{image}(T)$ with $\dim W$.
3. Before computing, use dimensions: $\dim V > \dim W$ rules out injectivity; $\dim V < \dim W$ rules out surjectivity.
4. Matrix of $T \circ S$: multiply $[T]_\beta^\gamma$ by $[S]_\alpha^\beta$, row by column.
5. Entrywise proofs: compare (i,j) entries using the definition of matrix product.

Worked examples in the notes

- $T(u_1, u_2) = (u_1,\ 2u_1 + u_2,\ -u_1 + 3u_2) \to$ columns $(1,2,-1)^T$ and $(0,1,3)^T \to [T]v = 0$ reduces to $u_1 = u_2 = 0 \to$ injective.
- $D : P_3(\mathbb{R}) \to P_2(\mathbb{R})$, $D(p) = p' \to [D]$ has pivot columns 2, 3, 4 $\to$ image basis $\{1, 2x, 3x^2\}$, dimension $3 = \dim P_2(\mathbb{R}) \to$ surjective (kernel dimension 1 by rank-nullity).

- No injective linear map $\mathbb{R}^{10} \to \mathbb{R}^2 \to \dim V > \dim W$; no surjective linear map $\mathbb{R}^2 \to \mathbb{R}^{10} \to \dim V < \dim W$.
- $\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 1 & 4 \end{pmatrix} = \begin{pmatrix} 2 & -4 \\ 1 & 4 \end{pmatrix} \to$ row-by-column $\to$ used to argue non-commutativity (the reverse product is a suggested exercise; scan partly unclear).

Common pitfalls

- For a composition TS to be defined, the codomain of the inner map must match the domain of the outer map.
- The "injective iff surjective" equivalence needs $\dim V = \dim W$; the notes stress this.
- Finiteness matters: in infinite dimensions a space can have a proper subspace of the same dimension, so the dimension test for surjectivity fails.
- Matrix multiplication is not commutative (suggested exercise using the example above).

Week 9: Inverses of Linear Transformations

Big idea

A function has an inverse exactly when it is bijective, and the inverse of a linear transformation is again linear. For linear maps between finite-dimensional spaces, the inverse is computed by row reducing the matrix of the map against the identity.

Definitions

- **Inverse of** $S : V \to W$: a function $T : W \to V$ with $T(S(\vec{v})) = \vec{v}$ for all $\vec{v} \in V$ and $S(T(\vec{w})) = \vec{w}$ for all $\vec{w} \in W$.
- **Isomorphism**: an invertible linear transformation $T : V \to W$.
- **Endomorphism**: a linear transformation $T : V \to V$ (linear operator); $\mathcal{L}(V)$ is the set of all of them.
- **Automorphism**: an invertible endomorphism; $\text{Aut}(V) \subseteq \mathcal{L}(V)$ is the set of these.
- **Isomorphic**: V and W are isomorphic if an isomorphism $T : V \to W$ exists.
- **Group**: $\text{Aut}(V)$ under composition is a group (associative, identity I, inverses).
- **Elementary row operations**: (i) add λ times row i to row j, with $i \neq j$; (ii) multiply a row by $\lambda \neq 0$; (iii) interchange two rows.

Theorems and results

- **Inverse exists iff bijective.** S has an inverse iff S is injective and surjective. *Idea:* an inverse gives surjectivity via $\vec{v} = T(\vec{w})$ and injectivity by applying T to $S(\vec{v}_1) = S(\vec{v}_2)$. Conversely, define $T(\vec{w}) = \vec{v}$ where $S(\vec{v}) = \vec{w}$.
- **Uniqueness.** An inverse, if it exists, is unique. *Idea:* every $\vec{w}$ equals $S(\vec{v})$ for some $\vec{v}$, so two inverses agree on $\vec{w}$.
- **Inverse of a linear map is linear.** If linear S has inverse T, then T is linear. *Idea:* write vectors as $S(\cdot)$ and use linearity of S.
- **Row operations are invertible.** Scaling by λ is undone by scaling by $\dfrac{1}{\lambda}$; $R_j + \lambda R_i$ is undone by $R_j - \lambda R_i$; a swap is its own inverse. Each operation is left-multiplication by an invertible elementary matrix E.
- **Inverse algorithm.** If $[M \mid I]$ reduces to $[A \mid M']$ in RREF with $A = I$, then $M^{-1} = M'$; otherwise M is not invertible. *Idea:* $E_k \cdots E_1 M = I$, so the product is a left inverse, hence the inverse.
- **Matrix of an inverse.** For invertible T and bases, the matrix of T^{-1} is the inverse of the matrix of T (exact index placement [unclear in scan]; proof left as exercise).
- **Isomorphism test.** $T : V \to W$ is an isomorphism iff it is injective and surjective. If $\dim V = \dim W$ is finite, either one suffices.
- **Dimension criterion.** Finite-dimensional real spaces V, W are isomorphic iff $\dim V = \dim W$. Any n-dimensional space over F is isomorphic to F^n.

Techniques / procedures

1. **Invert** M**:** form $[M \mid I]$, row reduce to RREF, and read M^{-1} from the right block if the left block is I.
2. **Invert** $T : V \to W$**:** pick bases α, β; compute $[T]$; invert the matrix; define T^{-1} from the inverse matrix.
3. **Prove an isomorphism:** check linearity; show $\ker T = \{\vec{0}\}$ (injective); show surjectivity, or use dimension equality to need only one.
4. **Show** $V \cong F^n$**:** choose a basis, map $\sum a_i \vec{v}_i \mapsto (a_1, \ldots, a_n)$, check linearity, injectivity and surjectivity.

Worked examples in the notes

- $T(a+bx) = (a+b, a)$, $P_1(\mathbb{R}) \to \mathbb{R}^2$, standard bases $\to [T] = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$, inverse matrix $\begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}$, so $T^{-1}(a,b) = b + (a-b)x$.
- $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix}$ $\to$ row reduces to I $\to$ invertible (inverse entries [unclear in scan]).
- $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 3 & 7 & 10 \end{pmatrix}$ $\to$ reduction yields a zero row $\to$ not invertible.
- $M_{2\times 2}(\mathbb{R}) \cong \mathbb{R}^4$ via $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto (a,b,c,d)$ $\to$ linear, ker $= \{\vec{0}\}$, surjective.
- $P_3(\mathbb{R}) \cong \mathbb{R}^4$ via $a_0 + a_1x + a_2x^2 + a_3x^3 \mapsto (a_0, a_1, a_2, a_3)$ $\to$ proof left as exercise.

Common pitfalls

- The scalar in $R_i \to \lambda R_i$ must be nonzero, or the operation cannot be undone.
- $R_j + \lambda R_i$ requires $i \neq j$.
- The inverse algorithm always works in theory, but it is numerically unstable and costly.

Week 10: Determinants and Change of Bases

Big idea

The determinant is the unique alternating, multilinear function of the rows of a square matrix with value 1 at I, and it decides invertibility. It also measures signed area and volume scaling and gives Cramer's rule for solving systems.

Definitions

- **Multilinear**: linear in each of its n inputs.
- **Alternating**: swapping two entries changes the sign of the output.
- **Diagonal matrix**: $D = [d_{ij}]$ with $d_{ij} = 0$ whenever $i \neq j$.
- ij**-minor** M_{ij}: the $(n-1) \times (n-1)$ matrix obtained by deleting row i and column j of M.
- **Determinant**: the alternating, multilinear function $\det : M_{n\times n}(\mathbb{R}) \to \mathbb{R}$ (inputs are rows $R_1, \dots, R_n$) with $\det(I) = 1$.
- n**-area /** n**-volume**: the size of the parallelotope spanned by n vectors. For $\vec{v}_1, \dots, \vec{v}_n \in \mathbb{R}^n$ it equals $|\det(\vec{v}_1, \dots, \vec{v}_n)|$.
- **Determinant of a linear map**: the determinant of any matrix representation of T.

Theorems and results

- **Dependent or repeated inputs.** An alternating multilinear function outputs 0 if two inputs are equal, and also if the inputs are linearly dependent. *Idea:* swapping equal entries gives $T = -T$, so $2T = 0$. For dependence, expand the redundant vector by multilinearity; every term has a repeated input.
- **Diagonal.** $\det D = d_{11}d_{22}\cdots d_{nn}$.
- **Row operations.** Scaling a row by λ multiplies det by λ. Swapping two rows multiplies det by -1. Adding a multiple of one row to another leaves det unchanged.
- **Invertibility.** M is invertible iff $\det M \neq 0$. If M is not invertible, its rows are dependent and $\det M = 0$. If M is invertible, Gauss–Jordan reduction gives $\det M \neq 0$.
- **Expansion.** For any fixed row i: $\det M = \sum_j (-1)^{i+j} m_{ij} \det M_{ij}$. The column version also holds, and every choice of row or column gives the same value.
- **Transpose.** $\det A = \det A^T$ (proof left as exercise).
- **Area scaling.** For $T : \mathbb{R}^n \to \mathbb{R}^n$, $|\det(T\vec{v}_1, \dots, T\vec{v}_n)| = |\det[T]| \cdot |\det(\vec{v}_1, \dots, \vec{v}_n)|$. In $\mathbb{R}^2$ the signed area of the parallelogram on (a, c) and (b, d) is $ad - bc$ (proved in the notes for a special case).
- **Cramer's rule.** If $\det A \neq 0$, the solution of $A\vec{x} = \vec{b}$ has $x_j = \dfrac{\det A_j}{\det A}$, where A_j is A with column j replaced by $\vec{b}$.

Techniques / procedures

1. **Row expansion:** choose a row, compute $\sum_j (-1)^{i+j} m_{ij} \det M_{ij}$, and recurse on the minors.
2. **Row reduction:** reduce M toward I, track the scalar each operation contributes, and combine them with $\det I = 1$.
3. **Small cases:** $\det[a] = a$ and $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$.
4. **Cramer's rule:** compute $\det A$; for each j form A_j; set $x_j = \det A_j / \det A$.
5. **Area of a parallelotope:** take $|\det|$ of the matrix whose columns are the edge vectors.

Worked examples in the notes

- det $\begin{pmatrix} 1 & 0 & -1 \\ 2 & 2 & 3 \\ 3 & 1 & 4 \end{pmatrix}$ $\to$ expand along row 1 $\to 5 + 4 = 9 \to$ nonzero, so invertible.
- det $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ $\to$ multilinearity on $\vec{e}_1, \vec{e}_2 \to ad - bc$.
- $T(x, y) = (x + 2y, 2x + y)$ (matrix as read from scan) $\to |\det| = |-3| = 3$, so the unit square maps to area 3.
- $T(x, y) = (x, 0) \to$ the square collapses to a segment $\to$ area 0 and det $= 0$.
- A 2×2 system solved by Cramer's rule $\to [x, y]^T = [0, -\frac{1}{2}]^T$ (coefficients partly [unclear in scan]).

Common pitfalls

- The determinant does change under row operations; "row operation invariance" is "not quite right", but the changes are simple and predictable.
- Row reduction and recursive expansion are costly and numerically unstable.
- Cramer's rule applies only when $\det A \neq 0$.
- The determinant of a linear map is computed from a matrix representation of it.

Week 11: Change of Basis and Eigenvalue and Eigenvectors

Big idea

Eigenvectors are nonzero vectors that a linear map only rescales, and a matrix similar to a diagonal matrix makes its powers and polynomials easy to compute. Change-of-basis matrices relate coordinates across bases, which is what lets eigenvectors produce a diagonalizing basis.

Definitions

- **Polynomial of a matrix**: for $p(x) = \sum_i a_i x^i$, $p(A) = \sum_i a_i A^i$ with $A^0 = I$.
- **Similar matrices**: $A, B \in M_{n\times n}(F)$ are similar if $B = M^{-1}AM$ for some invertible M.
- **Diagonalizable matrix**: a matrix similar to a diagonal matrix.
- **Eigenvector and eigenvalue**: for $M \in M_{n\times n}(F)$, a nonzero $\vec{x}$ with $M\vec{x} = \lambda\vec{x}$ for some $\lambda \in F$. For $T \in \mathcal{L}(V)$, a nonzero $\vec{v}$ with $T(\vec{v}) = \lambda\vec{v}$.
- **Eigenspace**: $E_\lambda = \{\vec{v} : T(\vec{v}) = \lambda\vec{v}\} = \ker(T - \lambda I)$, a subspace.
- **Characteristic polynomial**: $\chi_A(x) = \det(A - xI)$.
- **Change-of-basis matrix**: the matrix that converts coordinates from one basis to another, $[\vec{v}]_\beta = [I]\,[\vec{v}]_\alpha$ (index placement [unclear in scan]).

Theorems and results

- **Diagonal powers.** For diagonal D, $(D^k)_{ii} = d_{ii}^k$. Hence $p(D)$ is diagonal with entries $p(d_{ii})$ (induction, left as exercise).
- **Similarity and powers.** If $B = M^{-1}AM$, then $A^k = MB^kM^{-1}$ and $p(A) = Mp(B)M^{-1}$. *Idea:* the inner $M^{-1}M$ factors cancel.
- **Eigenvectors in coordinates.** $\vec{v}$ is an eigenvector of T with eigenvalue λ iff $[\vec{v}]_\alpha$ is an eigenvector of $[T]_\alpha$ with eigenvalue λ.
- **Kernel form.** $M\vec{x} = \lambda\vec{x}$ iff $\vec{x} \in \text{null}(M - \lambda I)$.
- **Eigenvalues are roots.** λ is an eigenvalue iff $\chi(\lambda) = 0$. *Idea:* $\det(A - \lambda I) = 0$ iff $A - \lambda I$ has a nonzero kernel vector.
- **Count.** An $n \times n$ matrix has at most n complex eigenvalues, and at most n real ones. *Idea:* χ has degree n, and the fundamental theorem of algebra factors it.
- **Basis independence.** The eigenvalues of T do not depend on the basis used to compute them.
- **Change of basis.** Change-of-basis matrices are inverse to each other. Matrix representations of T in two bases are related by change-of-basis matrices (exact placement [unclear in scan]).

Techniques / procedures

1. **Eigenpairs of a matrix:** (a) compute $\det(A - xI)$; (b) factor to find the roots λ_j; (c) row reduce $A - \lambda_j I$ and take nonzero solutions as eigenvectors.
2. **Eigenpairs of T:** pick a basis α, compute $[T]_\alpha$, apply the steps above, then convert eigenvectors back using α.
3. **Compute $p(A)$ for diagonalizable A:** find M, D with $A = MDM^{-1}$; compute $p(D)$ entrywise; return $p(A) = Mp(D)M^{-1}$.
4. **Change-of-basis matrix:** write the coordinates of each basis vector in the other basis as columns; invert by Gauss–Jordan when needed.

Worked examples in the notes

- $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \to$ direct multiplication $\to A^2 = \begin{pmatrix} 7 & 10 \\ 15 & 22 \end{pmatrix}$.
- $p(x) = 3x^2 + 2x + 5$ applied to $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \to 3A^2 + 2A + 5I \to \begin{pmatrix} 28 & 34 \\ 51 & 79 \end{pmatrix}$.
- Diagonal matrix with transcendental entries $\to$ powers computed entrywise.
- $T(x, y) = (2x + y, x + 2y) \to T(1, 1) = (3, 3) \to (1, 1)$ is an eigenvector with eigenvalue 3.
- $[T] = \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix} \to \chi = (\lambda - 5)(\lambda + 1) \to E_5 = \text{span}\{(1, 2)\}$, $E_{-1} = \text{span}\{(-1, 1)\}$. (The printed formula $(x + 4y, 2x + 3y)$ does not match this matrix [unclear in scan].)
- $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \to \chi = (1 - x)(2 - x)^2 \to$ eigenvalues 1 and 2.
- $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \to \chi = x^2 + 1 \to$ no real eigenvalues; eigenvalues $\pm i$ over $\mathbb{C}$.
- Fibonacci $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \to \chi = x^2 - x - 1 \to$ eigenvalues $\dfrac{1 \pm \sqrt{5}}{2}$, eigenvectors $t(\lambda, 1)$.

Common pitfalls

- The zero vector is never accepted as an eigenvector.
- Not every matrix has eigenvalues: they must lie in the field used, as the $\mathbb{R}$-rotation example shows.
- Only real roots count when working over $\mathbb{R}$.
- Finding eigenvalues and eigenvectors by hand is laborious, so follow the fixed sequence of steps.

Week 12: Diagonalizability

Big idea

A linear operator is diagonalizable exactly when V has a basis of eigenvectors. Distinct eigenvalues guarantee this, and in general algebraic and geometric multiplicities decide it.

Definitions

- **Diagonalizable operator**: $T \in \mathcal{L}(V)$ for which some basis α makes $[T]_\alpha$ diagonal, equivalently V has a basis of eigenvectors.
- **Diagonalizable matrix**: a matrix similar to a diagonal matrix, i.e. $A = PDP^{-1}$.
- **Algebraic multiplicity** $\text{Alg}(\lambda)$: the number of factors $(x - \lambda)$ in the linear factorization of the characteristic polynomial.
- **Geometric multiplicity** $\text{Geo}(\lambda) = \dim E_\lambda$.
- **Dependence relation**: a polynomial p with $p(T) = Z$, where Z is the zero map.

Theorems and results

- **Eigenbasis.** T is diagonalizable iff V has a basis of eigenvectors; the diagonal entries are then the eigenvalues. *Idea:* a diagonal matrix has columns $T(\vec{v}_i) = \lambda_i \vec{v}_i$, and conversely.
- **Matrix form.** A is diagonalizable iff A has n linearly independent eigenvectors. *Idea:* $AP = PD$ forces the columns of P to be eigenvectors.
- **Distinct eigenvalues.** Eigenvectors for distinct eigenvalues are linearly independent. *Idea:* take a shortest dependence, apply T, subtract a multiple of the original, and get a shorter dependence. Corollary: n distinct real eigenvalues on an n-dimensional real space give a diagonalizable operator.
- **Geometric $\leq$ algebraic.** For a real eigenvalue, $\text{Geo}(\lambda) \leq \text{Alg}(\lambda)$. *Idea:* extend a basis of E_λ to a basis of V; the characteristic polynomial then has a factor $(t - \lambda)^{\text{Geo}(\lambda)}$.
- **Eigenspaces.** For $\lambda_1 \neq \lambda_2$, $E_{\lambda_1} \cap E_{\lambda_2} = \{\vec{0}\}$. Independent sets from distinct eigenspaces combine into an independent set.
- **Criterion.** With distinct real eigenvalues λ_j of multiplicities m_j, T is diagonalizable iff $\dim V = \sum_j m_j$ and $\dim E_{\lambda_j} = m_j$ for each j. Equivalently, the eigenspaces sum to V.
- **Dependence relation.** If $p(T) = Z$ and λ is an eigenvalue of T, then $p(\lambda) = 0$. The roots of p are the only possible eigenvalues; not every root need be one.
- **Idempotent case.** If $T^2 = T$, then $V = E_0 + E_1$ with $E_0 \cap E_1 = \{\vec{0}\}$, so T is diagonalizable. *Idea:* $\vec{v} = (\vec{v} - T\vec{v}) + T\vec{v}$.
- **Matrix functions.** If $A = PDP^{-1}$, then $f(A) = Pf(D)P^{-1}$, with $f(D)$ computed entrywise. Likewise $e^A = Pe^D P^{-1}$, $\sin A = P \sin(D) P^{-1}$, $\cos A = P \cos(D) P^{-1}$.

Techniques / procedures

1. **Diagonalize A:** (a) find eigenvalues from χ_A; (b) solve $(A - \lambda I)\vec{x} = \vec{0}$ for each λ; (c) if n independent eigenvectors exist, set $P = [\vec{v}_1 \cdots \vec{v}_n]$ and $D = \text{diag}(\lambda_i)$.
2. **Test via multiplicities:** factor χ_A for Alg; row reduce $A - \lambda I$ and count free variables for Geo. Check Alg = Geo only when Alg > 1.
3. **Compute $f(A)$:** evaluate f on each diagonal entry of D, then return $Pf(D)P^{-1}$.
4. **Compute $f(T)$:** choose a basis α, form $A = [T]_\alpha$, diagonalize, compute $f(A)$, and convert back with $[f(T)]_\alpha = f(A)$.
5. **Eigenvector from a dependence relation:** apply the factors $(T - \lambda_i I)$ successively to a nonzero $\vec{v}$ until the result is $\vec{0}$; the last nonzero vector is an eigenvector for the last factor used.

Worked examples in the notes

- $T(x, y) = (x + 4y, 2x + 3y) \to$ eigenvalues $5, -1$, eigenvectors $(1, 1)$ and $(-2, 1)$, which are independent $\to$ diagonalizable.
- $A = \begin{pmatrix} 0 & 3 & 0 \\ 1 & 0 & -1 \\ 0 & 2 & 0 \end{pmatrix} \to \chi = -\lambda(\lambda - 1)(\lambda + 1) \to$ eigenvectors $(1, 0, 1)$, $(3, 1, 2)$, $(3, -1, 2)$ for $0, 1, -1 \to$ diagonalizable.
- $\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix} \to \text{Alg}(1) = 2$ but $\text{Geo}(1) = 1 \to$ the criterion fails.
- $\begin{pmatrix} 7 & -4 & 0 \\ 8 & -5 & 0 \\ 6 & -6 & 3 \end{pmatrix} \to \chi = -(\lambda - 3)^2(\lambda + 1)$; $E_3 = \text{span}\{(1, 1, 0), (0, 0, 1)\}$, so $\text{Geo}(3) = \text{Alg}(3) = 2 \to$ diagonalizable.
- $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \to$ dependence $A^2 - A - I = 0 \to$ possible eigenvalues are the roots of $x^2 - x - 1$.
- $A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} \to$ dependence $(A - 5I)(A + I) = 0 \to$ possible eigenvalues $5, -1$; the vector $(2, 3)$ is an eigenvector for neither.

Common pitfalls

- The zero vector is never an eigenvector.
- The roots of a dependence relation are only candidate eigenvalues; not all of them need be eigenvalues.
- Alg = Geo must be verified only for eigenvalues with Alg > 1.

Homework 12: Diagonalizability and Applying Polynomials

Recall list

1. Diagonal Matrix
2. Similar Matrices
3. Diagonalizable Matrix
4. Conditions for Diagonalizability
5. Theorem: A Matrix is Diagonalizable iff it has n Linearly Independent Eigenvectors
6. Linearly Independent Eigenvectors
7. Theorem: Distinct Eigenvalues $\Rightarrow$ Diagonalizable
8. Algebraic Multiplicity
9. Geometric Multiplicity
10. Theorem: Algebraic vs. Geometric Multiplicity

Problem groups

1. Matrix diagonalization (A_1 to A_5). Topic: diagonalizability of 2×2 and 3×3 matrices (A_1 diagonal, A_2 upper triangular with repeated eigenvalue, A_3 diagonal with a repeated entry, $A_4 = \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix}$, A_5 upper triangular). Task: decide whether each is diagonalizable; for each diagonalizable one, construct an invertible P and diagonal D with $A = PDP^{-1}$.

2. Diagonalizable linear operators (T_1 to T_6). Topic: operators on $\mathbb{R}^2$, P_2 and $M_{2\times 2}$, including $T_1(x, y) = (2x, 3y)$, $T_2(x, y) = (4x + y, 4y)$, $T_3(p) = p'$, $T_4(p) = xp(x)$, $T_5(A) = A^T$, and T_6 (multiplication by a fixed matrix [side unclear in scan]). Task: decide diagonalizability; for each diagonalizable operator, construct a basis β with $[T]_\beta$ diagonal, and the change-of-basis matrices $[I]_\beta^\alpha$ and $[I]_\alpha^\beta$, where α is the standard basis.

3. Applying polynomials to matrices. Topic: $f(x) = 3x^2+2x-5$ applied to the diagonalizable matrices from group 1. Task: compute $f(A)$ for each, using $f(A) = Pf(D)P^{-1}$.

4. Applying polynomials to operators. Topic: $f(x) = x^2+2x-5$ applied to the diagonalizable operators from group 2. Task: explicitly define $f(T)$ for each.

5. Proofs (beyond the final's scope). Topic: (a) the characteristic polynomial $\chi_A(x)$ of $A \in M_{n\times n}(F)$; (b) the equivalence $AP = PD \iff A\vec{v}_i = \lambda_i \vec{v}_i$ for all i, where $P = [\vec{v}_1 \cdots \vec{v}_n]$ and D is diagonal. Task: prove each statement.